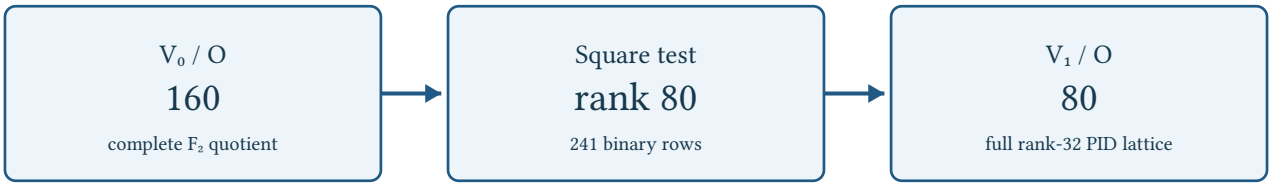
R65: first Frobenius lattice step

The complete trace-dual quotient $\frac{V_0}{O}$ has binary dimension 160. The first native run squares every class using the full rank-32 source multiplication table. Complete denominator remainders produce 241 binary rows of rank 80, leaving 80 survivors. Every remainder, kernel product, and whole-field survivor is checked. An altered kernel is rejected.

First complete descending Frobenius step



Exact algebraic dimensions; runs first-frobenius65-01 and lattice65-01. No timing comparison.

The second native run reconstructs V_1 as a rank-32 polynomial lattice. It checks both module inclusions, both basis inverse products, all 32 basis squares back in V_0, and an altered-inverse negative control. The exact binary quotient lengths are 80 for $\frac{V_1}{O}$ and 80 for $\frac{V_0}{V_1}$. This is one descending step; equality with the integral closure requires iteration to a stable lattice.

Execution and evidence

Both attempts finished with native exit 0, exact markers, and zero Singular error lines. The first and second UTC wall intervals are 76 s and 187 s, respectively; these are receipts, not controlled performance comparisons. Twelve historical source/protocol/driver byte pins and the saved quotient, kernel, survivor, and lattice bundles are recorded in results.json. The input field bridge is R64, while the Gram and multiplication table are the retained R61 and R52 sources. The next full test is $V_2 = \{x \in V_1 : x^2 \in V_1\}$ on every class of $\frac{V_1}{O}$.

The normalization method follows Singh–Swanson, Theorem 1.1.